

Activity - Estimating Lifetime

From `6_regression.ipynb` : build a Kaplan-Meier survival analysis on the Telco churn data and compare lifetime between payment methods.

The sections follow the ten-step workflow in Kim & Kim, *Survival Analysis 101: An Easy Start Guide to Analyzing Time-to-Event Data*, European Journal of Cardiovascular Nursing.

```
In [1]: from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from lifelines import KaplanMeierFitter
from lifelines.plotting import add_at_risk_counts
from lifelines.statistics import multivariate_logrank_test,
pairwise_logrank_test
from lifelines.utils import restricted_mean_survival_time

pd.set_option("display.precision", 3)

df = pd.read_csv("Telco-Churn.csv")
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

1. Identify the event

The event is churn: the customer cancels the service. `Churn` records it as yes/no, and there is only one kind of ending in this data - nobody dies, moves abroad or gets transferred to another product - so there are no competing risks to separate out.

2. Determine the follow-up duration

`tenure` is the follow-up clock: months elapsed since the customer joined, counted up to the moment the data was collected.

```

In [2]: print(f"tenure: {df['tenure'].min()} to {df['tenure'].max()} months, "
        f"{df['tenure'].nunique()} distinct values")
        print(f"one row per customer: {df['customerID'].nunique()} ids across {len(df)} rows")

```

```

tenure: 0 to 72 months, 73 distinct values
one row per customer: 7043 ids across 7043 rows

```

So every survival curve below stops at **month 72** - a six-year window with no month missing inside it, and no customer counted twice.

3. Select the time scale

Months, because that is the only scale the data offers - `tenure` arrives already rounded to whole months, so a finer scale cannot be recovered.

4. Organize the data structure

A Kaplan-Meier fit needs three things per customer: how long they were observed, whether that time ended in the event, and which group they are in.

survival term	Telco column	why this one
duration T	<code>tenure</code>	the only column measuring elapsed time, in months
event E	<code>Churn == "Yes"</code>	whether the time ended in churn, or is still running
group	<code>PaymentMethod</code>	what the activity asks to compare
-	<code>customerID</code>	not a variable, just a check that one row means one customer

The other seventeen are customer attributes, and a Kaplan-Meier curve has nowhere to put them.

```
In [3]: df["T"] = df["tenure"].astype(float)
df["E"] = (df["Churn"] == "Yes").astype(int)

WORKING = ["customerID", "PaymentMethod", "tenure", "Churn", "T", "E"]
df[WORKING].head()
```

```
Out[3]:
```

	customerID	PaymentMethod	tenure	Churn	T	E
0	7590-VHVEG	Electronic check	1	No	1.0	0
1	5575-GNVDE	Mailed check	34	No	34.0	0
2	3668-QPYBK	Mailed check	2	Yes	2.0	1
3	7795-CFOCW	Bank transfer (automatic)	45	No	45.0	0
4	9237-HQITU	Electronic check	2	Yes	2.0	1

5. Declare the survival data

In Stata this is the `stset` command. lifelines has no such step - every fitter takes the durations and the event indicator as arguments - so what belongs here instead is the one

row type that cannot be declared at all: a customer observed for zero months.

```
In [4]: print(f"tenure == 0: {(df['T'] == 0).sum()} rows (signed up too recently to be
observed)")

# A zero-length observation never enters the risk set for any month, so it
carries no
# information for the estimator.
df = df[df["T"] > 0].copy()
print(f"kept {len(df)} rows")
```

```
tenure == 0: 11 rows (signed up too recently to be observed)
kept 7032 rows
```

6. Examine descriptive statistics

The point of Kaplan-Meier: most rows are *censored*, not observed failures. What that costs if it is ignored:

```
In [5]: print(f"customers          {len(df)}")
print(f"churned (events) {df.E.sum()} ({df.E.mean():.1%})")
print(f"still active      {(1 - df.E).sum()} (right-censored)")
print()

overall_km = KaplanMeierFitter().fit(df["T"], df["E"])

print(f"drop the censored rows, average the rest "
      f"{df.loc[df.E == 1, 'T'].mean():5.1f} months")
print(f"average tenure, as if everyone had ended {df['T'].mean():5.1f}
months")
print(f"Kaplan-Meier, RMST over the 72-month window "
      f"{restricted_mean_survival_time(overall_km, t=72):4.1f} months")
```

```
customers          7032
churned (events) 1869  (26.6%)
still active      5163  (right-censored)
```

```
drop the censored rows, average the rest    18.0 months
average tenure, as if everyone had ended    32.4 months
Kaplan-Meier, RMST over the 72-month window 54.5 months
```

The first number keeps only customers who left, so every survivor is discarded. The second keeps them but files each one as finished today. Both answer a question nobody asked, and both land far below the estimate that uses the censored rows for what they do say - that the customer lasted *at least* this long.

Life table

`event_table` is the life table the workflow asks for. Each row is one month: how many customers were at risk, how many churned, and how many left the risk set without

churning because the data ran out on them.

```
In [6]: overall_km.event_table.head(12)
```

```
Out[6]:
```

	removed	observed	censored	entrance	at_risk
event_at					
0.0	0	0	0	7032	7032
1.0	613	380	233	0	7032
2.0	238	123	115	0	6419
3.0	200	94	106	0	6181
4.0	176	83	93	0	5981
5.0	133	64	69	0	5805
6.0	110	40	70	0	5672
7.0	131	51	80	0	5562
8.0	123	42	81	0	5431
9.0	119	46	73	0	5308
10.0	116	45	71	0	5189
11.0	99	31	68	0	5073

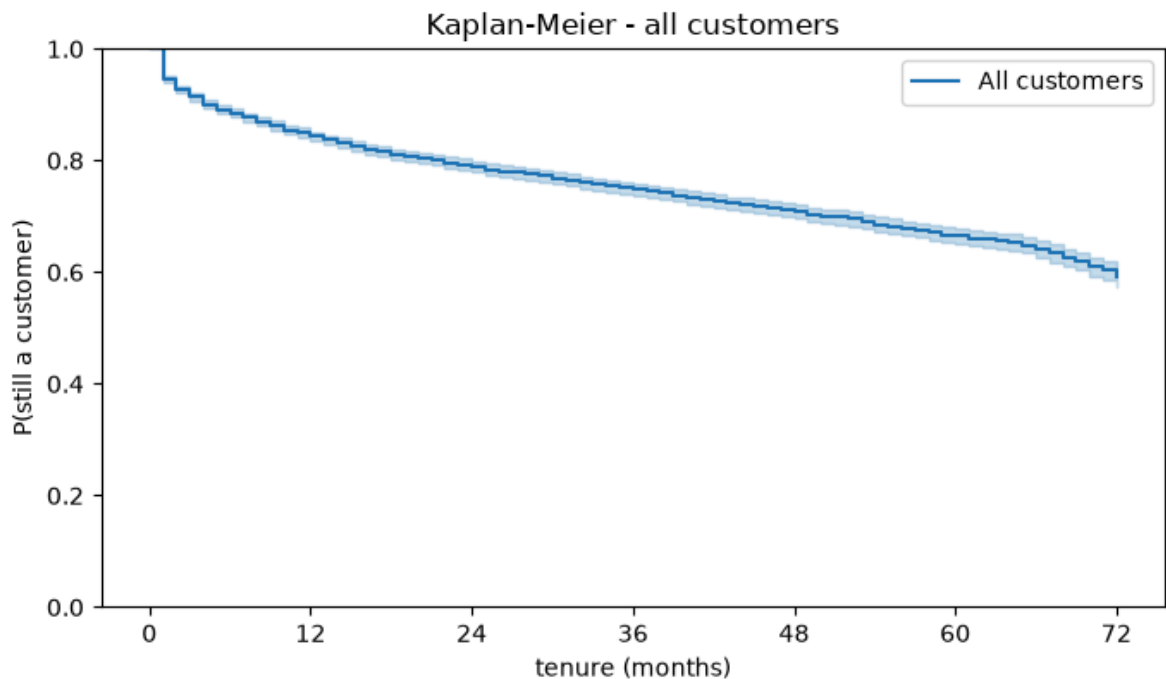
7. Estimate the survival curve

```
In [7]: YEAR_TICKS = list(range(0, 73, 12)) # a tick per year, still counted in months

kmf = KaplanMeierFitter().fit(df["T"], event_observed=df["E"], label="All
customers")

ax = kmf.plot_survival_function(figsize=(7, 4.2))
ax.set(xlabel="tenure (months)", ylabel="P(still a customer)",
       title="Kaplan-Meier - all customers", ylim=(0, 1), xticks=YEAR_TICKS)
plt.tight_layout()
plt.show()

print("median survival time:", kmf.median_survival_time_)
print(kmf.survival_function_at_times([12, 24, 48, 72]).round(3))
```



```
median survival time: inf
12    0.843
24    0.789
48    0.709
72    0.593
Name: All customers, dtype: float64
```

8. Compare the groups

One curve per payment method, with the number still at risk printed underneath and the log-rank result on the panel - the reporting checklist the paper gives for a survival graph.

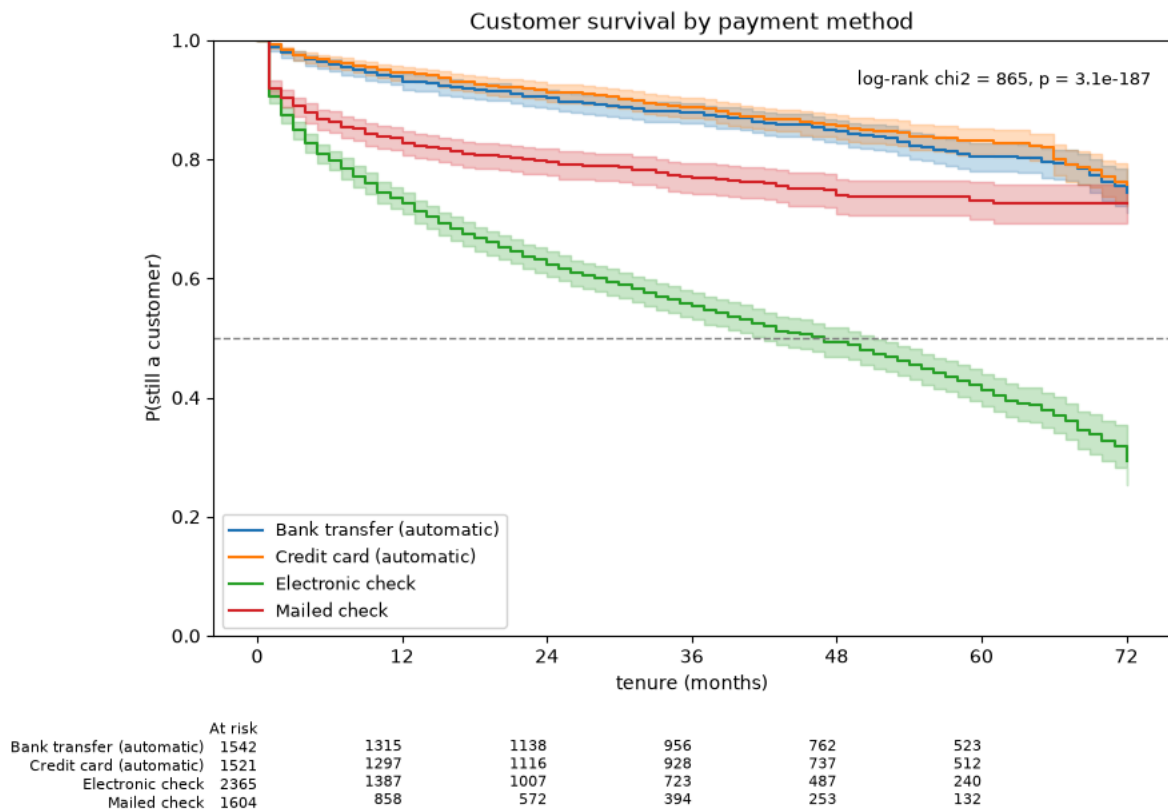
```
In [8]: METHODS = sorted(df["PaymentMethod"].unique())
fitters = {}

fig, ax = plt.subplots(figsize=(9, 6.5))
for method in METHODS:
    group = df[df["PaymentMethod"] == method]
    f = KaplanMeierFitter().fit(group["T"], event_observed=group["E"],
    label=method)
    fitters[method] = f
    f.plot_survival_function(ax=ax, ci_show=True)

overall = multivariate_logrank_test(df["T"], df["PaymentMethod"], df["E"])
ax.set(xlabel="tenure (months)", ylabel="P(still a customer)",
    title="Customer survival by payment method", ylim=(0, 1),
    xticks=YEAR_TICKS)
ax.axhline(0.5, color="grey", lw=1, ls="--")
ax.legend(loc="lower left", fontsize=9)
ax.text(0.98, 0.95, f"log-rank chi2 = {overall.test_statistic:.0f}, p =
{overall.p_value:.1e}",
    transform=ax.transAxes, ha="right", va="top", fontsize=9)
```

```
# lifelines counts those at risk *after* each tick, so month 72 - the last
month anyone
# was observed - would read zero for every group and is left off the table.
add_at_risk_counts(*fitters.values(), ax=ax, fontsize=8,
                  rows_to_show=["At risk"], xticks=YEAR_TICKS[:-1])

plt.tight_layout()
Path("figures").mkdir(exist_ok=True)
fig.savefig("figures/km_by_payment_method.png", dpi=150, bbox_inches="tight")
plt.show()
```



Is the difference real?

The log-rank test compares the curves without assuming any shape for them.

```
In [9]: print(f"all four groups: chi2 = {overall.test_statistic:.1f}, p =
{overall.p_value:.3g}")
print()

pairwise = pairwise_logrank_test(df["T"], df["PaymentMethod"], df["E"])
pairwise.summary[["test_statistic", "p"]].round(4)
```

all four groups: chi2 = 865.2, p = 3.07e-187

Out[9]:

		test_statistic	p
Bank transfer (automatic)	Credit card (automatic)	0.868	0.351
	Electronic check	510.035	0.000
	Mailed check	51.072	0.000
Credit card (automatic)	Electronic check	539.740	0.000
	Mailed check	64.815	0.000
Electronic check	Mailed check	152.456	0.000

How much lifetime separates them

`median_survival_time_` is the statistic the lecture uses, but three of the four curves never reach 0.5 inside the 72 months observed, so it comes back as `inf` for them. Two statistics that do exist for every group:

- **survival at fixed times** - the share still subscribed at 12, 24, 48, 72 months
- **RMST** (restricted mean survival time) - the area under the curve up to a chosen horizon, read as "expected months as a customer during the first N"

RMST needs that horizon picked by hand, and the choice is not neutral: over a short window everybody survives most of it, so the groups look closer together. Reporting several horizons keeps that visible. Twelve months is the one to quote - the counts under the graph above are still deep there, where by month 72 they are down to a handful.

```
In [10]: HORIZONS = [12, 24, 48, 72]
HEADLINE = 12

rows = []
for method, f in fitters.items():
    s = f.survival_function_at_times(HORIZONS).to_numpy()
    row = {
        "n": int(f.event_observed.shape[0]),
        "churned": int(f.event_observed.sum()),
        "median": f.median_survival_time_,
    }
    row |= {f"S({h})": v for h, v in zip(HORIZONS, s)}
    row |= {f"RMST({h})": restricted_mean_survival_time(f, t=h) for h in
HORIZONS}
    rows.append({"PaymentMethod": method} | row)

summary =
pd.DataFrame(rows).set_index("PaymentMethod").sort_values(f"RMST({HEADLINE})")
summary
```


Out[10]:

	n	churned	median	S(12)	S(24)	S(48)	S(72)	RMST(12)	RMST(24)
PaymentMethod									
Electronic check	2365	1071	47.0	0.727	0.624	0.493	0.295	9.863	17.9
Mailed check	1604	308	inf	0.828	0.797	0.740	0.726	10.545	20.2
Bank transfer (automatic)	1542	258	inf	0.931	0.902	0.847	0.745	11.561	22.5
Credit card (automatic)	1521	232	inf	0.945	0.913	0.855	0.758	11.623	22.7

In [11]:

```
# How much of each horizon does a customer of this group actually spend
# subscribed?
share = pd.DataFrame({
    f"{h} mo": summary[f"RMST({h})"] / h for h in HORIZONS
})
share.loc["gap (best - worst)"] = share.max() - share.min()
share
```

Out[11]:

	12 mo	24 mo	48 mo	72 mo
PaymentMethod				
Electronic check	0.822	0.748	0.653	0.573
Mailed check	0.879	0.845	0.808	0.783
Bank transfer (automatic)	0.963	0.941	0.909	0.875
Credit card (automatic)	0.969	0.949	0.918	0.886
gap (best - worst)	0.147	0.201	0.265	0.313

9. Multivariate model - Cox proportional hazards

The curves above compare payment methods without holding anything else constant, and payment method is not chosen at random - it travels with contract type.

In [12]:

```
pd.crosstab(df["PaymentMethod"], df["Contract"], normalize="index").round(3)
```

Out[12]:

	Contract	Month-to-month	One year	Two year
PaymentMethod				
Bank transfer (automatic)		0.382	0.254	0.364
Credit card (automatic)		0.357	0.262	0.381
Electronic check		0.782	0.147	0.071
Mailed check		0.557	0.209	0.234

Electronic-check customers are on month-to-month contracts 78% of the time against 36% for credit card, so the lifetime gap above could be contract type in disguise. A Cox model holds contract constant and separates the two.

The coefficients are read as **hazard ratios**, `exp(coef)` : the multiplier on the instantaneous churn rate relative to the reference level, at every point in time. HR = 2 means twice the churn rate of the reference; HR = 1 means no difference.

Reference levels: credit card (automatic), two-year contract.

```
In [13]: from lifelines import CoxPHFitter
from lifelines.statistics import proportional_hazard_test

REFERENCE = ["PaymentMethod_Credit card (automatic)", "Contract_Two year"]

design = pd.get_dummies(df[["PaymentMethod", "Contract"]],
dtype=float).drop(columns=REFERENCE)
design["tenure"] = df["T"].to_numpy()
design["churned"] = df["E"].to_numpy()

payment_cols = [c for c in design.columns if c.startswith("PaymentMethod")]
design.head()
```

Out[13]:

	PaymentMethod_Bank transfer (automatic)	PaymentMethod_Electronic check	PaymentMethod_Mailed check	Contract_1 to
0	0.0	1.0	0.0	
1	0.0	0.0	1.0	
2	0.0	0.0	1.0	
3	1.0	0.0	0.0	
4	0.0	1.0	0.0	

Model A - payment method alone

```
In [14]: cox_a = CoxPHFitter().fit(design[payment_cols + ["tenure", "churned"]],
    "tenure", "churned")

    cox_a.summary[["exp(coef)", "exp(coef) lower 95%", "exp(coef) upper 95%", "p"]]
```

Out[14]:

	exp(coef)	exp(coef) lower 95%	exp(coef) upper 95%	p
covariate				
PaymentMethod_Bank transfer (automatic)	1.089	0.912	1.301	3.446e-01
PaymentMethod_Electronic check	4.621	4.005	5.332	1.130e-97
PaymentMethod_Mailed check	2.149	1.810	2.551	2.540e-18

Model B - payment method adjusted for contract type

```
In [15]: cox_b = CoxPHFitter().fit(design, "tenure", "churned")

    print(f"concordance model A {cox_a.concordance_index_:.3f}    model B  

    {cox_b.concordance_index_:.3f}")
    cox_b.summary[["exp(coef)", "exp(coef) lower 95%", "exp(coef) upper 95%", "p"]]
```

concordance model A 0.676 model B 0.809

Out[15]:

	exp(coef)	exp(coef) lower 95%	exp(coef) upper 95%	p
covariate				
PaymentMethod_Bank transfer (automatic)	1.028	0.861	1.228	7.593e-01
PaymentMethod_Electronic check	2.104	1.822	2.430	3.981e-24
PaymentMethod_Mailed check	1.848	1.555	2.197	3.447e-12
Contract_Month-to-month	54.102	39.712	73.706	3.566e-141
Contract_One year	6.888	4.944	9.598	4.068e-30

```
In [16]: comparison = pd.DataFrame({
    "HR alone (A)": cox_a.hazard_ratios_,
    "HR adjusted (B)": cox_b.hazard_ratios_[payment_cols],
})
    comparison["shrinkage"] = 1 - comparison["HR adjusted (B)"] / comparison["HR
```

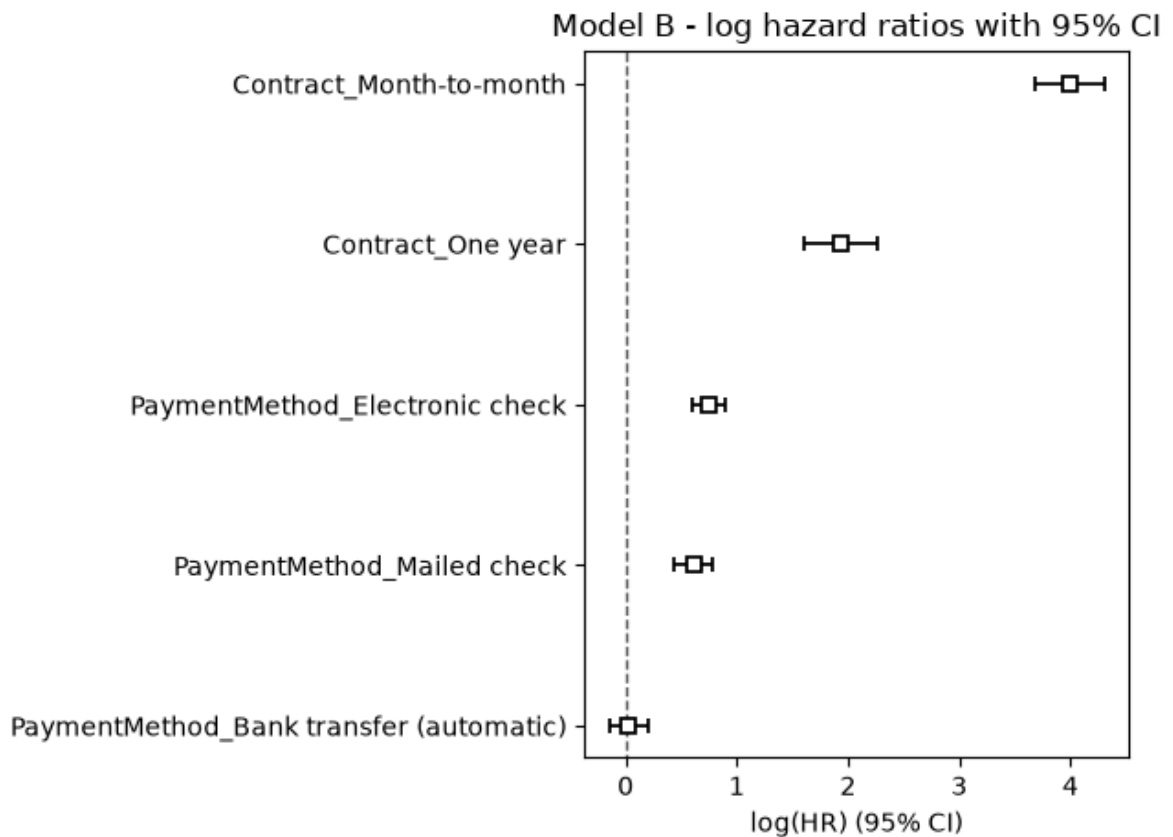
```
alone (A)"]  
comparison
```

Out[16]:

	HR alone (A)	HR adjusted (B)	shrinkage
covariate			
PaymentMethod_Bank transfer (automatic)	1.089	1.028	0.056
PaymentMethod_Electronic check	4.621	2.104	0.545
PaymentMethod_Mailed check	2.149	1.848	0.140

In [17]:

```
ax = cox_b.plot()  
ax.set_title("Model B - log hazard ratios with 95% CI")  
plt.tight_layout()  
plt.savefig("figures/cox_hazard_ratios.png", dpi=150, bbox_inches="tight")  
plt.show()
```



Does the model's own assumption hold?

Cox assumes a covariate's hazard ratio is *constant over time*. That is testable, and here it fails for some covariates - so each HR above should be read as an average effect across the 72 months, not a fixed multiplier. The mailed-check curve in section 8 shows why: it drops steeply in the first months and then flattens.

```
In [18]: proportional_hazard_test(cox_b, design,
time_transform="rank").summary[["test_statistic", "p"]]
```

```
Out[18]:
```

	test_statistic	p
Contract_Month-to-month	20.686	5.411e-06
Contract_One year	0.008	9.286e-01
PaymentMethod_Bank transfer (automatic)	0.303	5.821e-01
PaymentMethod_Electronic check	9.082	2.582e-03
PaymentMethod_Mailed check	30.595	3.180e-08

10. Interpret and report

Minimum follow-up time

Two different questions hide under "how long do we need to watch customers".

How soon can a conclusion be drawn? Censor every customer administratively at month `t` - as if the study had stopped there - and re-run the log-rank test. The smallest `t` whose test comes back significant is the minimum follow-up that particular comparison needs.

How long do the estimates stay trustworthy? A survival curve is only as good as the number of customers still at risk behind it. Once the risk set thins out, the curve is carried by a handful of people and the confidence band widens accordingly.

```
In [19]: FOLLOWUPS = range(1, 73)
T_all, E_all, G_all = df["T"].to_numpy(), df["E"].to_numpy(),
df["PaymentMethod"].to_numpy()

first_significant = {}
for t in FOLLOWUPS:
    censored_T = np.minimum(T_all, t)
    censored_E = ((E_all == 1) & (T_all <= t)).astype(int)
    for pair, p in pairwise_logrank_test(censored_T, G_all,
censored_E).summary["p"].items():
        if pair not in first_significant and p < 0.05:
            first_significant[pair] = (t, p)

full = pairwise_logrank_test(T_all, G_all, E_all).summary["p"]
rows = []
for pair in full.index:
    t, p = first_significant.get(pair, (np.nan, np.nan))
    rows.append({"pair": " vs ".join(pair),
                "min follow-up (months)": t,
                "p at that point": p,
```

```

        "p on full 72 months": full[pair]])

pd.DataFrame(rows).sort_values("min follow-up (months)").set_index("pair")

```

Out[19]:

	min follow-up (months)	p at that point	p on full 72 months
pair			
Bank transfer (automatic) vs Electronic check	1.0	5.673e-25	6.231e-113
Bank transfer (automatic) vs Mailed check	1.0	1.930e-19	8.905e-13
Credit card (automatic) vs Electronic check	1.0	1.765e-28	2.148e-119
Credit card (automatic) vs Mailed check	1.0	4.866e-23	8.226e-16
Electronic check vs Mailed check	2.0	5.579e-03	5.038e-35
Bank transfer (automatic) vs Credit card (automatic)	NaN	NaN	3.514e-01

In [20]:

```

at_risk = pd.DataFrame({
    method: [(group["T"] >= t).mean() for t in (12, 24, 36, 48, 60, 72)]
    for method, group in df.groupby("PaymentMethod")
}, index=[f"t={t}" for t in (12, 24, 36, 48, 60, 72)]).T
at_risk.columns.name = "fraction still at risk"
at_risk

```

Out[20]:

	fraction still at risk	t=12	t=24	t=36	t=48	t=60	t=72
Bank transfer (automatic)	0.871	0.748	0.627	0.507	0.355	0.104	
Credit card (automatic)	0.863	0.748	0.621	0.500	0.347	0.095	
Electronic check	0.605	0.440	0.311	0.210	0.113	0.017	
Mailed check	0.553	0.372	0.252	0.165	0.087	0.011	

Findings

In [21]:

```

best, worst = summary.index[-1], summary.index[0]

print(f"longest-lived group {best}")
print(f"shortest-lived group {worst}")
print()
for h in HORIZONS:
    b, w = summary.loc[best, f"RMST({h})"], summary.loc[worst, f"RMST({h})"]
    print(f"over {h:>2} months: {b:5.2f} vs {w:5.2f} -> gap {b - w:5.2f}
months ({(b - w) / h:.1%} of the window)")

```

```
print()
print(f"electronic-check hazard ratio, unadjusted
{cox_a.hazard_ratios_['PaymentMethod_Electronic check']:.2f}")
print(f"electronic-check hazard ratio, given contract
{cox_b.hazard_ratios_['PaymentMethod_Electronic check']:.2f}")
```

longest-lived group Credit card (automatic)
shortest-lived group Electronic check

over 12 months: 11.62 vs 9.86 -> gap 1.76 months (14.7% of the window)
over 24 months: 22.78 vs 17.95 -> gap 4.82 months (20.1% of the window)
over 48 months: 44.05 vs 31.32 -> gap 12.73 months (26.5% of the window)
over 72 months: 63.81 vs 41.24 -> gap 22.57 months (31.3% of the window)

electronic-check hazard ratio, unadjusted 4.62
electronic-check hazard ratio, given contract 2.10

Lifetime by payment method. Over a first year, a credit-card customer spends 11.6 of the 12 months subscribed against 9.9 for an electronic-check customer - 97% of the window versus 82%. The two automatic methods cannot be told apart at all: the log-rank test gives $p = 0.35$ on the full data, and it never falls below 0.05 at any follow-up horizon. Mailed check sits between them. Median survival time, the statistic the lecture uses, only exists for electronic check (47 months) because the other three curves never fall to 0.5, so RMST does the comparing here.

The gap grows with the window. It is 1.8 months over a year, 12.7 months over four years, 22.6 months over the full six - and as a share of the window it climbs from 15% to 31%. So a single horizon flatters or exaggerates depending on where it is cut. Twelve months is the honest one to quote: 55-87% of each group is still at risk there, against 1-10% by month 72, so the longer figures are real but rest on a thinning tail.

How much of that is the payment method itself. On its own, electronic check carries a hazard ratio of 4.6 against credit card. Once contract type is held constant it falls to about 2.1 - so roughly half the apparent effect was contract type wearing a payment method's clothes. Mailed check ends up in the same place, an adjusted 1.85, but almost none of its effect was contract type to begin with: it shrinks 14% against electronic check's 55%. Both manual methods carry roughly double the churn rate of an automatic one once contract is held constant. Contract type is the only thing this model holds constant, though, so anything else that separates manual from automatic payers - income, age, how much someone trusts a standing order - is still free to be doing the work.

Minimum follow-up. Every comparison the data can support is already significant after **2 months** of follow-up: four of the five pairs separate within the first month, and electronic check versus mailed check needs a second. Watching longer sharpens the estimates but changes no conclusion - except for automatic bank transfer versus credit card, which stays indistinguishable however long anyone waits. At the other end, the two manual-payment

groups are down to 9-11% still at risk by month 60, so the tail of those curves should not be quoted.